

**Notes: Schmid, Valaitis & Villa (JFE, 2026)**  
**Government Debt Management and Inflation with Real and Nominal Bonds**

## Contents

|          |                                                                                                                                                             |           |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Big picture</b>                                                                                                                                          | <b>3</b>  |
| <b>2</b> | <b>Incremental contribution of the paper</b>                                                                                                                | <b>3</b>  |
| <b>3</b> | <b>Data and measurement / calibration</b>                                                                                                                   | <b>3</b>  |
| 3.1      | U.S. TIPS share . . . . .                                                                                                                                   | 3         |
| 3.2      | Calibration moments . . . . .                                                                                                                               | 4         |
| 3.3      | Policy moments . . . . .                                                                                                                                    | 4         |
| <b>4</b> | <b>Model and theoretical specification</b>                                                                                                                  | <b>4</b>  |
| 4.1      | Government portfolio . . . . .                                                                                                                              | 4         |
| 4.2      | Sticky prices and inflation . . . . .                                                                                                                       | 4         |
| 4.3      | Full Commitment . . . . .                                                                                                                                   | 5         |
| 4.4      | No Commitment . . . . .                                                                                                                                     | 5         |
| 4.5      | Portfolio tradeoff . . . . .                                                                                                                                | 5         |
| <b>5</b> | <b>Identification and theoretical design</b>                                                                                                                | <b>5</b>  |
| 5.1      | Full Commitment versus No Commitment . . . . .                                                                                                              | 5         |
| 5.2      | Single-bond economies as mechanism tests . . . . .                                                                                                          | 6         |
| 5.3      | No-lending constraint . . . . .                                                                                                                             | 6         |
| 5.4      | Long-term debt extension . . . . .                                                                                                                          | 6         |
| <b>6</b> | <b>Tables and figures</b>                                                                                                                                   | <b>6</b>  |
| 6.1      | Figure 1: U.S. TIPS share; Table 1: Parameter values . . . . .                                                                                              | 6         |
| 6.2      | Figure 2: Date two inflation policy functions; Figure 3: Date one optimal portfolio and corresponding inflationary bias . . . . .                           | 7         |
| 6.3      | Figure 4: Date two inflation and taxes; Figure 5: Date two inflation and inflationary bias with only nominal debt . . . . .                                 | 7         |
| 6.4      | Table 2: Full Commitment moments; Table 3: No Commitment moments . . . . .                                                                                  | 8         |
| 6.5      | Figure 6: No commitment dynamics; Table 4: No-lending policy moments . . . . .                                                                              | 8         |
| 6.6      | Table 5: No commitment and portfolio shares; Figure 7: Inflation and nominal-real portfolio rebalancing . . . . .                                           | 9         |
| 6.7      | Table 6: Comparative moments by average maturity of nominal debt; Table 7: Comparative moments by joint average maturity of nominal and TIPS debt . . . . . | 9         |
| <b>7</b> | <b>Short summary</b>                                                                                                                                        | <b>10</b> |
| <b>8</b> | <b>Conclusion</b>                                                                                                                                           | <b>10</b> |
| 8.1      | Core conclusion . . . . .                                                                                                                                   | 10        |
| 8.2      | Economic intuition . . . . .                                                                                                                                | 10        |

|     |                                              |    |
|-----|----------------------------------------------|----|
| 8.3 | Incremental contribution revisited . . . . . | 11 |
| 8.4 | Policy implication . . . . .                 | 11 |
| 8.5 | Limitations and open questions . . . . .     | 11 |
| 8.6 | Takeaway . . . . .                           | 11 |

# 1 Big picture

- **Question.** Can governments use real bonds such as Treasury Inflation-Protected Securities (TIPS) to tame inflation when nominal debt can be monetized ex post?
- **Core setting.** The government issues state-uncontingent nominal debt and real debt in an economy with sticky prices, distortionary taxes, fiscal expenditure shocks, and a monetary authority reacting to inflation.
- **Two regimes.** The paper compares Full Commitment, where the government can commit to future fiscal, inflation, and debt policies, with No Commitment, where successive governments respond strategically to inherited debt.
- **Main mechanism.** Nominal bonds provide insurance because inflation makes their real payoff state-contingent. Real bonds provide commitment because they cannot be inflated away.
- **Full Commitment result.** With full commitment, the government uses nominal debt for hedging and accumulates real assets. If government saving is ruled out, real bonds have little role.
- **No Commitment result.** Without commitment, the government issues real bonds to limit future governments' incentives to inflate away nominal debt. This generates a moderate and stable TIPS share similar to U.S. data.
- **Policy implication.** Raising the real-debt share reduces the inflationary temptation embedded in inherited nominal liabilities and can effectively curb inflation.
- **Economic takeaway.** Government debt composition is a fiscal-monetary commitment device. Real bonds are valuable not only for hedging inflation risk but also for disciplining future monetary/fiscal policy.

# 2 Incremental contribution of the paper

- **Adds real bonds to optimal debt management with sticky prices.** The paper studies government portfolios containing both nominal and inflation-indexed bonds in a New Keynesian-style environment.
- **Separates hedging from commitment.** Prior debt-management work emphasizes fiscal hedging with nominal debt. This paper shows that real bonds can be useful because they reduce the temptation to monetize nominal debt.
- **Contrasts Full Commitment and No Commitment quantitatively.** The paper does not only solve a Ramsey benchmark. It solves Markov-perfect/time-consistent policies and shows that the no-commitment equilibrium better matches the U.S. TIPS share.
- **Explains the small but positive TIPS share.** The model rationalizes why TIPS are present but not dominant in the U.S. debt portfolio: real bonds mitigate commitment problems, but nominal bonds remain valuable for fiscal insurance.
- **Links debt maturity, indexed debt, and inflation bias.** The paper extends the model to long-term nominal and real debt and shows how maturities alter inflation incentives.
- **Provides a fiscal-monetary interpretation of TIPS.** TIPS are not just investor inflation-protection securities. In the model they are a policy tool that changes the incentives of future governments.

# 3 Data and measurement / calibration

## 3.1 U.S. TIPS share

The motivating empirical fact is the low and persistent share of TIPS in U.S. marketable Treasury debt. Figure 1 plots the TIPS face value share from the Treasury monthly statement of public debt. The share rises after introduction, reaches roughly 10 percent around the late 2000s, and then stabilizes around a high-single-digit percentage.

**Interpretation:** The low-but-positive TIPS share is the target fact. A model in which TIPS are useless cannot explain the positive share. A model in which TIPS dominate cannot explain why the share remains modest.

### 3.2 Calibration moments

The calibration uses annual U.S. data on government expenditures, labor tax rates, and debt moments. Government spending is modeled as an exogenous stochastic process. Inflation dynamics are pinned down by a monetary policy rule and a New Keynesian Phillips curve.

- Government expenditure averages about 20 percent of GDP.
- Steady-state annual inflation is set to 2 percent.
- The Taylor rule response to inflation is calibrated at 1.2 in the baseline.
- Preferences, price rigidity, price elasticity, labor elasticity, and adjustment costs are calibrated to match standard macro moments.

**Interpretation:** The calibration is designed to make the debt-management problem quantitative: fiscal shocks, labor taxes, and inflation all move at magnitudes close to U.S. experience.

### 3.3 Policy moments

The paper reports unconditional moments from simulated equilibrium dynamics: average inflation, average tax rate, real and nominal debt shares, correlations among fiscal variables and inflation, inflation volatility, and debt persistence.

**Interpretation:** These moments allow the paper to compare theoretical debt portfolios with the observed TIPS share and with the observed stability of inflation and debt.

## 4 Model and theoretical specification

### 4.1 Government portfolio

The government can issue nominal debt  $B_t$  and real debt  $b_t$ . Nominal debt is repaid in dollars, so its real payoff is affected by inflation. Real debt is indexed and cannot be inflated away. A simplified representation of the government budget tradeoff is:

$$\text{tax revenue} + \text{new debt issuance} = \text{spending} + \text{debt service}.$$

The real value of nominal repayment falls when inflation rises, while the real value of TIPS repayment does not.

**Interpretation:** This difference is the whole reason the portfolio matters. Nominal bonds are useful for state-contingent fiscal hedging; real bonds are useful for commitment.

### 4.2 Sticky prices and inflation

Inflation is costly because prices are sticky and inflation distortions are real. Monetary policy follows a Taylor-style rule, and inflation also feeds into the real return on nominal debt.

$$i_t = \phi_\pi \pi_t + \text{other terms}.$$

**Interpretation:** A stronger monetary response to inflation reduces the temptation to use inflation for fiscal purposes, thereby changing the optimal amount of real debt.

### 4.3 Full Commitment

Under Full Commitment, the planner chooses fiscal, inflation, and debt policies for all future dates. Because it internalizes all future tradeoffs, it can exploit inflation to make nominal debt state-contingent and smooth fiscal shocks.

- Nominal debt is valuable because it can be diluted when expenditure shocks are high.
- Real bonds are less attractive because they do not provide this inflation-hedging channel.
- In the unconstrained full-commitment model, the government borrows nominally and saves in real bonds.

**Interpretation:** With commitment, the planner uses debt composition for insurance. It can promise future inflation choices and therefore exploit nominal debt without creating a time-consistency problem.

### 4.4 No Commitment

Under No Commitment, the current government commits to repay debt but cannot commit future governments to future taxes, inflation, or debt choices. Future governments may be tempted to inflate away inherited nominal debt.

large inherited nominal debt  $\Rightarrow$  future temptation to inflate.

Households anticipate this and demand compensation when buying nominal debt.

**Interpretation:** The current government issues more real debt because real debt reduces the inflation incentive faced by future governments. This lowers inflation risk and borrowing costs.

### 4.5 Portfolio tradeoff

The optimal portfolio balances two forces:

1. **Insurance:** nominal debt gives the government state-contingent real payoffs through inflation.
2. **Incentives:** real debt limits the temptation to inflate inherited nominal liabilities.

**Interpretation:** The observed TIPS share emerges from this tradeoff. Too little real debt leaves future inflation temptation high. Too much real debt sacrifices the hedging role of nominal debt.

## 5 Identification and theoretical design

This is a structural/theoretical paper rather than a reduced-form causal paper. The identification-style logic is the comparison of equilibria across commitment regimes and debt-market restrictions.

### 5.1 Full Commitment versus No Commitment

The main theoretical comparison holds preferences, technology, fiscal shocks, and monetary policy fixed while changing only the government's ability to commit to future policies.

**Interpretation:** If the optimal portfolio changes sharply across the two regimes, the difference can be attributed to the commitment friction rather than to changes in fiscal risk or preferences.

## 5.2 Single-bond economies as mechanism tests

The paper studies environments in which the government can issue only real bonds or only nominal bonds.

- With real bonds only, inflation is anchored but tax smoothing is limited.
- With nominal bonds only, taxes can be smoothed, but inflationary incentives are severe.
- With both instruments, the government can combine insurance and commitment.

**Interpretation:** These counterfactuals isolate why each instrument matters. Real debt disciplines inflation; nominal debt hedges fiscal shocks.

## 5.3 No-lending constraint

The paper also restricts the government from saving in private bonds. This matters because the unconstrained full-commitment portfolio can involve leveraged nominal debt and negative real debt. The no-lending constraint makes the model more realistic and changes the role of real bonds.

**Interpretation:** This test shows that some full-commitment portfolio results depend on allowing the government to take large asset positions. The no-commitment case remains closer to observed portfolios.

## 5.4 Long-term debt extension

The long-term extension allows nominal and real bonds with different average maturities. Longer nominal maturity changes the inflation incentive because current and expected future inflation affect the value of outstanding liabilities.

**Interpretation:** The extension shows that debt maturity and debt indexation interact. Longer nominal maturity can strengthen the inflationary dilution motive, increasing the value of TIPS as a commitment tool.

# 6 Tables and figures

## 6.1 Figure 1: U.S. TIPS share; Table 1: Parameter values

**Read:** Figure 1 shows the empirical target: the TIPS share is positive but modest and persistent. Table 1 reports the calibrated preference, firm, monetary-policy, and government-spending parameters.

**Economic message:** The figure motivates the empirical puzzle, while the table pins down the quantitative environment used to explain it.

**Interpretation:** The key interpretation is that a realistic model must not predict either zero TIPS or a fully indexed debt portfolio. It must rationalize a stable middle ground.

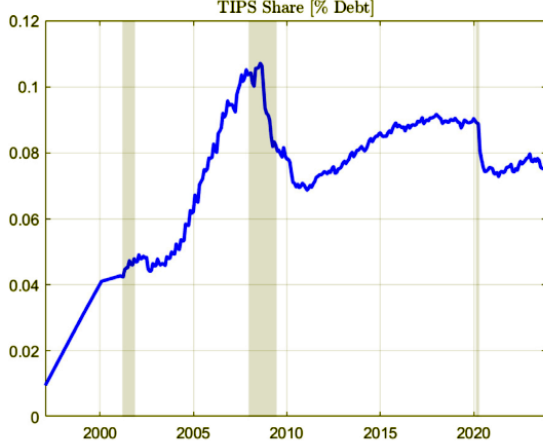


Figure 1: U.S. TIPS share.

**Table 1**  
Parameter values.

|                 |                         | Parameter  | Value  |
|-----------------|-------------------------|------------|--------|
| Preferences     | Discount factor         | $\beta$    | 0.96   |
|                 | Risk aversion           | $\eta_c$   | 2      |
|                 | Labor disutility        | $\chi$     | 4.3276 |
|                 | Labor elasticity        | $\eta_l$   | 1.8    |
| Firm            | Price elasticity        | $1/\nu$    | 10     |
|                 | Adjustment cost         | $\varphi$  | 20     |
|                 | Response to inflation   | $\phi_\pi$ | 1.2    |
| Monetary Policy | Average $g$             | $\mu_g$    | 0.068  |
|                 | Volatility of $\log(g)$ | $\sigma_g$ | 0.016  |
|                 | Autocorr. of $\log(g)$  | $\rho_g$   | 0.977  |

*Notes:* The table reports the parameter values used in the quantitative part of the paper. The same calibration is used in the Full Commitment model without TIPS bonds and in all extensions in presented in Online Appendix B.1.2 and in the No Commitment model in Section 5.3.

Table 1: Parameter values.

## 6.2 Figure 2: Date two inflation policy functions; Figure 3: Date one optimal portfolio and corresponding inflationary bias

**Read:** Figure 2 shows how date-two inflation responds to nominal and real debt in low- and high-spending states. Figure 3 shows the date-one portfolio choice and the corresponding strategic inflation-bias terms.

**Economic message:** Together these figures show how nominal debt creates inflation incentives and how portfolio composition shapes the strategic interaction across governments.

**Interpretation:** Figure 2 displays the ex-post temptation: more nominal debt raises the value of inflating. Figure 3 shows the ex-ante response: the government tilts the portfolio toward real debt when commitment is limited.

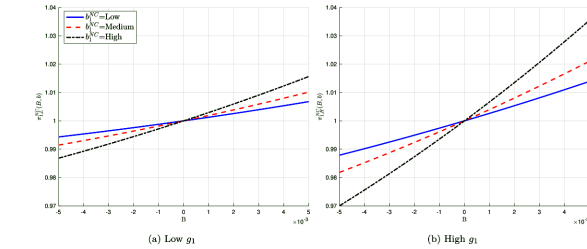


Figure 2: Date two inflation policy functions.

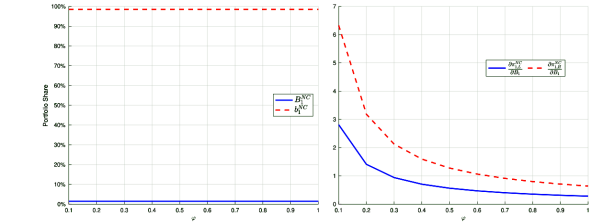


Figure 3: Date one optimal portfolio and corresponding inflationary bias.

## 6.3 Figure 4: Date two inflation and taxes; Figure 5: Date two inflation and inflationary bias with only nominal debt

**Read:** Figure 4 plots inflation and taxes in the two-bond no-commitment economy. Figure 5 repeats the exercise in a nominal-bond-only economy.

**Economic message:** Removing real bonds makes inflation and inflationary bias much larger. Real debt disciplines the future government because it cannot be inflated away.

**Interpretation:** The difference between the figures is the paper’s clearest mechanism test: TIPS reduce the ex-post incentive to monetize debt, while the nominal-only economy relies too much on inflation.

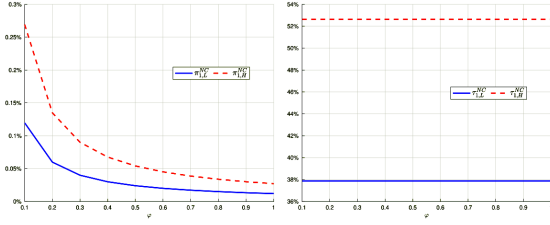


Figure 4: Date two inflation and taxes.

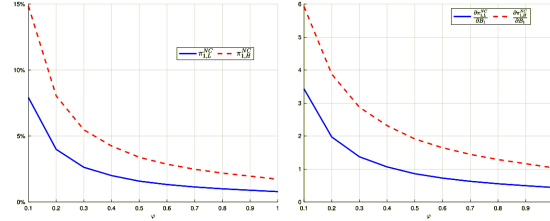


Figure 5: Nominal-debt-only inflation and inflationary bias.

## 6.4 Table 2: Full Commitment moments; Table 3: No Commitment moments

**Read:** Table 2 summarizes full-commitment policy moments with and without TIPS. Table 3 summarizes no-commitment moments with and without TIPS.

**Economic message:** Under full commitment, real bonds play a limited or even negative portfolio role. Under no commitment, real bonds emerge as a commitment device and reduce inflation.

**Interpretation:** The tables show why the no-commitment model better matches the data: it delivers a positive real-debt share and relatively stable inflation.

of government expenditure shocks. The simulation is initialized at  $b = B = 0$  and we drop the first 100 periods before calculating moments.

**Table 2**

Summary of moments: Policy variables.

| Description                 | Moments                | FC No TIPS | FC baseline |
|-----------------------------|------------------------|------------|-------------|
| Avg. Inflation [%]          | $E(\pi) - 1$           | 2.0        | 1.8         |
| Avg. Tax [%]                | $E(\tau)$              | 22.0       | 23.2        |
| Avg. Real to GDP            | $E(b/Y)$               | –          | –0.2        |
| Avg. Nominal to GDP         | $E(B/Y)$               | –0.2       | 0.3         |
| Autocorr. Real Share        | $\rho_1(b/(b+B))$      | –          | 0.046       |
| Corr. Gov. Spending and GDP | $\rho(g, Y)$           | 0.587      | 0.822       |
| Corr. Tax and GDP           | $\rho(\tau, Y)$        | –0.434     | 0.441       |
| Corr. Inflation and GDP     | $\rho(\pi, Y)$         | –0.245     | 0.667       |
| Corr. Inflation and Real    | $\rho(\pi, b)$         | –          | 0.455       |
| Corr. Inflation and Nominal | $\rho(\pi, B)$         | 0.74       | –0.588      |
| Std. Inflation              | $100\sigma(\log(\pi))$ | 0.033      | 0.164       |
| Autocorr Inflation          | $\rho_1(\pi)$          | 0.6268     | 0.8981      |
| Std. Debt                   | $100\sigma(\log(b+B))$ | 16.3840    | 4.2275      |
| Autocorr Debt               | $\rho_1(b+B)$          | 0.9999     | 0.9984      |

Notes: The table reports sample moments from simulating the equilibrium dynamics of Table 2: Summary of full-commitment moments.

**Table 3**

Summary of moments: Policy variables.

| Description                     | Moments                | NC No TIPS | NC Baseline |
|---------------------------------|------------------------|------------|-------------|
| Avg. Inflation [%]              | $E(\pi) - 1$           | 1.67       | 1.93        |
| Avg. Tax [%]                    | $E(\tau)$              | 26.0       | 25.81       |
| Avg. Real to GDP                | $E(b/Y)$               | –          | 0.65        |
| Avg. Nominal to GDP             | $E(B/Y)$               | 0.30       | –1.06       |
| Autocorr. Real Portfolio Weight | $\rho_1(b/(b+B))$      | –          | 0.899       |
| Corr. Gov. Spending and GDP     | $\rho(g, Y)$           | 0.99       | 1.00        |
| Corr. Tax and GDP               | $\rho(\tau, Y)$        | 0.953      | 0.998       |
| Corr. Inflation and GDP         | $\rho(\pi, Y)$         | –0.719     | 0.932       |
| Corr. Inflation and Real        | $\rho(\pi, b)$         | –          | 0.946       |
| Corr. Inflation and Nominal     | $\rho(\pi, B)$         | 0.975      | –0.808      |
| Std. Inflation                  | $100\sigma(\log(\pi))$ | 0.141      | 0.056       |
| Autocorr Inflation              | $\rho_1(\pi)$          | 0.954      | 0.944       |
| Std. Debt                       | $100\sigma(\log(b+B))$ | 0.481      | 0.121       |
| Autocorr Debt                   | $\rho_1(b+B)$          | 0.978      | 0.950       |

Notes: The table reports sample moments of the No Commitment model with and without TIPS bonds from simulating model equilibrium dynamics using the same

Table 3: Summary of no-commitment moments.

## 6.5 Figure 6: No commitment dynamics; Table 4: No-lending policy moments

**Read:** Figure 6 plots the dynamic response to an expenditure shock under no commitment with and without TIPS. Table 4 compares full commitment and no commitment under no-lending constraints.

**Economic message:** TIPS dampen inflation responses after fiscal shocks. The no-lending constraint makes the comparison more realistic and highlights why real debt can appear in observed portfolios.

**Interpretation:** In the baseline with TIPS, the inflation response is smaller and the debt portfolio adjusts more smoothly. Without TIPS, the government relies more on inflation and taxes.



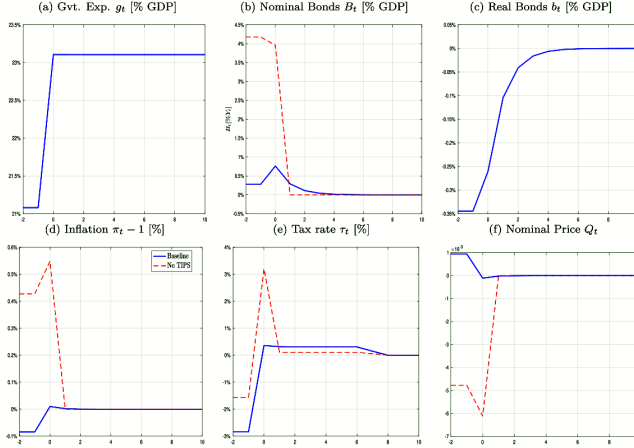


Figure 6: No commitment conditional dynamics.

commitment: conditional dynamics after an expenditure shock. The figure displays the dynamics of fiscal variables around a shock that increases government spending, at  $t = 0$ . X-axis report time  $t$ . Solid blue line: baseline model with TIPS bonds. Dashed red line: benchmark model without TIPS bonds. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

**Table 4**  
Summary of moments: Policy variables.

| Description                 | Moments                 | FC     | FC No Lend | NC     | NC No Lend |
|-----------------------------|-------------------------|--------|------------|--------|------------|
| Avg. Inflation [%]          | $E(\pi_t - 1)$          | 1.8    | 1.97       | 1.93   | 1.67       |
| Avg. Tax [%]                | $E(\tau_t)$             | 23.2   | 25.3       | 25.8   | 25.4       |
| Avg. Real Share             | $E(b_t/(b_t + B_t))$    | -0.5   | 0          | -1.6   | 0.07       |
| Avg. Nominal Share          | $E(B_t/(B_t + b_t))$    | 1.403  | 0.998      | 2.6    | 0.93       |
| Autocorr. Real Share        | $\rho(b_t/(b_t + B_t))$ | 0.046  | 0.957      | 0.899  | 0.935      |
| Corr. Gov. Spending and GDP | $\rho(g_t, Y)$          | 0.822  | 0.777      | 1.00   | 0.997      |
| Corr. Tax and GDP           | $\rho(\tau_t, Y)$       | 0.441  | 0.366      | 0.998  | 0.988      |
| Corr. Inflation and GDP     | $\rho(\pi_t, Y)$        | 0.607  | 0.545      | 0.932  | 0.505      |
| Corr. Inflation and Real    | $\rho(\pi_t, b)$        | 0.455  | -0.071     | 0.946  | -0.719     |
| Corr. Inflation and Nominal | $\rho(\pi_t, B)$        | -0.588 | 0.644      | -0.808 | 0.679      |

Note: The table reports sample moments from simulating model equilibrium dynamics using the same realization of government expenditure.

Table 4: No-lending moments.

## 6.6 Table 5: No commitment and portfolio shares; Figure 7: Inflation and nominal-real portfolio rebalancing

**Read:** Table 5 varies monetary-policy and inflation-cost parameters and reports the resulting real-debt share. Figure 7 asks how inflation changes when the portfolio is tilted toward real debt.

**Economic message:** A larger TIPS share lowers inflation. The effect is stronger when the amount of portfolio rebalancing is larger and when the debt-to-GDP ratio is high.

**Interpretation:** This pair gives the policy punchline: raising real debt can curb inflation because it reduces the value of inflating away inherited nominal liabilities.

**Table 5**  
No commitment and portfolio shares.

| Description                 | Moments                 | Model                                   |                                         |                                         |                                          |                                          |
|-----------------------------|-------------------------|-----------------------------------------|-----------------------------------------|-----------------------------------------|------------------------------------------|------------------------------------------|
|                             |                         | $\phi_2 = 1.2$<br>$\varphi = 20$<br>(a) | $\phi_2 = 1.2$<br>$\varphi = 15$<br>(b) | $\phi_2 = 1.2$<br>$\varphi = 10$<br>(c) | $\phi_2 = 1.17$<br>$\varphi = 20$<br>(d) | $\phi_2 = 1.15$<br>$\varphi = 20$<br>(e) |
| Avg. Inflation [%]          | $E(\pi_t - 1)$          | 1.67                                    | 1.57                                    | 1.11                                    | 1.58                                     | 1.53                                     |
| Avg. Tax [%]                | $E(\tau_t)$             | 25.45                                   | 25.81                                   | 25.82                                   | 25.41                                    | 25.40                                    |
| Avg. Real Share             | $E(b_t/(b_t + B_t))$    | 0.07                                    | 0.22                                    | 0.55                                    | 0.23                                     | 0.53                                     |
| Avg. Nominal Share          | $E(B_t/(B_t + b_t))$    | 0.93                                    | 0.78                                    | 0.45                                    | 0.77                                     | 0.47                                     |
| Autocorr. Real Share        | $\rho(b_t/(b_t + B_t))$ | 0.935                                   | 0.906                                   | 0.886                                   | 0.958                                    | 0.958                                    |
| Corr. Gov. Spending and GDP | $\rho(g_t, Y)$          | 0.997                                   | 0.995                                   | 1.00                                    | 1.00                                     | 1.00                                     |
| Corr. Tax and GDP           | $\rho(\tau_t, Y)$       | 0.988                                   | 0.974                                   | 0.998                                   | 0.999                                    | 0.999                                    |
| Corr. Inflation and GDP     | $\rho(\pi_t, Y)$        | 0.505                                   | -0.710                                  | 0.793                                   | 0.985                                    | 0.973                                    |
| Corr. Inflation and Real    | $\rho(\pi_t, b)$        | -0.719                                  | -0.726                                  | -0.241                                  | -0.716                                   | -0.822                                   |
| Corr. Inflation and Nominal | $\rho(\pi_t, B)$        | 0.679                                   | 0.820                                   | 0.245                                   | 0.716                                    | 0.822                                    |

Notes: The table reports the average inflation, taxes, portfolio weights of real and nominal bonds, and salient correlations among monetary and fiscal policy instruments. The simulation is initialized at  $t = 0 = 0$  and runs across the first 100 periods before calculating moments. All moments are calculated using the first 100 periods.

Table 5: No commitment and portfolio shares.

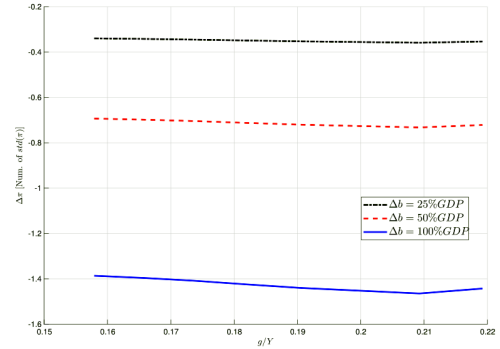


Figure 7: Inflation and nominal-real rebalancing.

## 6.7 Table 6: Comparative moments by average maturity of nominal debt; Table 7: Comparative moments by joint average maturity of nominal and TIPS debt

**Read:** Table 6 varies the average maturity of nominal debt. Table 7 varies both nominal and TIPS maturities jointly.

**Economic message:** Longer nominal debt strengthens future inflation incentives; longer TIPS strengthen discipline and permit lower inflation.

**Interpretation:** The maturity extension shows that the TIPS role depends not only on how much real debt is issued, but also on how long the real protection lasts.

**Table 6**  
Comparative moments by average maturity of nominal debt.

| Description                   | Moment                             | Avg. maturity of nominal debt |       |       |
|-------------------------------|------------------------------------|-------------------------------|-------|-------|
|                               |                                    | 1Y                            | 2Y    | 5Y    |
| Avg. Inflation (%)            | $\mathbb{E}(\pi - 1) \cdot 100$    | 1.67                          | 1.54  | 1.35  |
| Avg. Tax Rate (%)             | $\mathbb{E}(\tau) \cdot 100$       | 25.45                         | 28.07 | 37.42 |
| Std. Dev. of Inflation (%)    | $\sigma(\log \pi) \cdot 100$       | 0.07                          | 0.05  | 0.00  |
| Std. Dev. of Log Tax Rate (%) | $\sigma(\log(1 + \tau)) \cdot 100$ | 1.42                          | 1.28  | 0.93  |
| Share of Real Debt            | $\mathbb{E}(b^N / (b^N + B^N))$    | 0.07                          | 0.46  | 0.77  |
| Debt-to-GDP Ratio             | $\mathbb{E}((B^N + b^N)/Y)$        | 0.33                          | 0.36  | 0.64  |
| Real Debt-to-GDP Ratio        | $\mathbb{E}(b^N/Y)$                | 0.02                          | 0.17  | 0.49  |

*Notes:* This table reports simulated model moments under four different values of the average maturity of nominal debt, corresponding to different values of the parameter  $\delta_B$ . The columns report results for maturities of 1, 2 and 5 years, which correspond respectively to  $\delta_B = \{1, 0.5, 0.2\}$ . Reported moments include the average inflation rate, average tax rate, standard deviations of log inflation and log tax rate, the share of real debt in total debt, and the total and real debt-to-GDP ratios. The simulation is initialized at  $b = B = 0$  and we drop the first 100 periods before calculating moments. All moments are calculated in a simulation with  $T = 10000$  periods.

Table 6: Nominal debt maturity moments.

Comparative moments by joint average maturity of nominal and TIPS debt.

| Description                   | Moment                             | Avg. debt maturity |       |       |
|-------------------------------|------------------------------------|--------------------|-------|-------|
|                               |                                    | 1Y                 | 2Y    | 5Y    |
| Avg. Inflation (%)            | $\mathbb{E}(\pi - 1) \cdot 100$    | 1.67               | 1.40  | 1.25  |
| Avg. Tax Rate (%)             | $\mathbb{E}(\tau) \cdot 100$       | 25.45              | 31.51 | 47.28 |
| Std. Dev. of Inflation (%)    | $\sigma(\log \pi) \cdot 100$       | 0.07               | 0.09  | 0.00  |
| Std. Dev. of Log Tax Rate (%) | $\sigma(\log(1 + \tau)) \cdot 100$ | 1.42               | 1.12  | 0.73  |
| Share of Real Debt            | $\mathbb{E}(b^N / (b^N + B^N))$    | 0.07               | 0.47  | 0.55  |
| Debt-to-GDP Ratio             | $\mathbb{E}((B^N + b^N)/Y)$        | 0.33               | 0.40  | 0.47  |
| Real Debt-to-GDP Ratio        | $\mathbb{E}(b^N/Y)$                | 0.02               | 0.19  | 0.26  |

*Notes:* This table reports simulated model moments under three configurations in which the average maturity of nominal debt and TIPS coincide (1, 2 or 5 years). Reported moments include the average inflation rate, average tax rate, standard deviations of log inflation and log tax rate, the share of real debt in total debt, and the total and real debt-to-GDP ratios. The simulation is initialized at  $b = B = 0$  and we drop the first 100 periods before calculating moments. All moments are calculated in a simulation with  $T = 10000$  periods.

Table 7: Joint nominal and TIPS maturity moments.

## 7 Short summary

- The paper studies optimal government debt management with both nominal and real bonds.
- Nominal bonds are useful because inflation makes their real payoff state-contingent.
- Real bonds are useful because they cannot be inflated away, thereby reducing inflation incentives.
- Under full commitment, the government mainly uses nominal debt for fiscal hedging and may save in real bonds.
- Under no commitment, the government uses real debt as a commitment device.
- The no-commitment model delivers a moderate positive TIPS share and relatively stable inflation, matching U.S. experience better.
- Raising the TIPS share reduces the inflationary impact of fiscal shocks and inherited nominal debt.
- Longer debt maturities change the strength of the inflation incentive and the value of indexed debt.

## 8 Conclusion

### 8.1 Core conclusion

Schmid, Valaitis, and Villa show that the composition of government debt matters for inflation dynamics. Real bonds such as TIPS can serve as a commitment device because they cannot be inflated away. When governments lack commitment to future fiscal and monetary policies, issuing real debt reduces future incentives to monetize nominal liabilities.

### 8.2 Economic intuition

Nominal debt has an insurance benefit because inflation can reduce its real value in high-spending states. But that same feature creates an inflationary temptation for future governments. TIPS remove that temptation but also remove some fiscal-insurance value. The optimal portfolio balances these forces.

### **8.3 Incremental contribution revisited**

The paper's key contribution is to explain why real bonds can be valuable even when governments prefer nominal debt for hedging. The new force is limited commitment: TIPS discipline future governments and lower inflation incentives. This helps reconcile theory with the observed small but persistent U.S. TIPS share.

### **8.4 Policy implication**

A higher TIPS share can curb inflation by reducing the stock of nominal liabilities that future governments may want to inflate away. Debt management is therefore part of fiscal-monetary policy, not just a financing operation.

### **8.5 Limitations and open questions**

The model abstracts from some institutional details of Treasury issuance, heterogeneous investors, liquidity premia, and political constraints. It also treats fiscal and monetary commitment in a stylized way. Future work could connect the model to observed maturity choices, TIPS liquidity premia, and cross-country indexed-debt policies.

### **8.6 Takeaway**

The main takeaway is that inflation-indexed debt can be a commitment technology. A government that cannot credibly promise not to inflate nominal debt away can reduce that temptation by issuing more real debt.